PREMIUM

Full-Length Mock Test

Mock 03

Electrical Engineering

GATE EE

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Electrical Machines — Transformers]

No-load current in transformer is called:

A) Magnetizing current
B) Excitation current
C) No-load current
D) All of the above

Q.2 [1 Mark] [EASY] [Power Systems — Economic Dispatch]

Surge impedance loading (SIL) =:

A) V^2/Zc
B) V/Zc
C) Zc/V
D) V×Zc

Q.3 [1 Mark] [EASY] [Power Electronics — Inverters]

Full-wave bridge rectifier PIV:

A) V_m
B) $2V_m$
C) $V_m/2$
D) " 2 V_m

Q.4 [1 Mark] [EASY] [Control Systems — Compensators]

Ideal op-amp input impedance:

A) Zero
B) Infinite
C) 1M:•
D) Output impedance

Q.5 [1 Mark] [EASY] [Signals & Systems — Sampling]

Slip in induction motor is:

A) Ns/Nr
B) (Ns-Nr)/Ns
C) Nr/Ns
D) (Nr-Ns)/Nr

Q.6 [1 Mark] [EASY] [Analog Electronics — Op-Amps]

DC machine with lap winding has brushes:

A) 1
B) P
C) P+1
D) 2

Q.7 [1 Mark] [EASY] [Electrical Machines — DC Machines]

No-load current in transformer is called:

A) Magnetizing current

B) Excitation current

C) No-load current

D) All of the above

Q.8 [1 Mark] [EASY] [Power Systems — Load Flow]

Surge impedance loading (SIL) =:

A) V^2/Zc
B) V/Zc
C) Zc/V
D) V×Zc

Q.9 [1 Mark] [EASY] [Power Electronics — Inverters]

Full-wave bridge rectifier PIV:

A) V_m
B) 2V_m
C) V_m/2
D) " 2 V_m

Q.10 [1 Mark] [EASY] [Control Systems — Compensators]

Ideal op-amp input impedance:

A) Zero
B) Infinite
C) 1M:•
D) Output impedance

Q.11 [1 Mark] [EASY] [Signals & Systems — Convolution]

Slip in induction motor is:

A) Ns/Nr
B) (Ns-Nr)/Ns
C) Nr/Ns
D) (Nr-Ns)/Nr

Q.12 [1 Mark] [EASY] [Analog Electronics — BJT/FET Amplifiers]

DC machine with lap winding has brushes:

A) 1
B) P
C) P+1
D) 2

Q.13 [2 Marks] [MODERATE] [Electrical Machines — Servo Motors]

No-load current in transformer is called:

A) Magnetizing current
B) Excitation current
C) No-load current
D) All of the above

Q.14 [2 Marks] [MODERATE] [Power Systems — Economic Dispatch]

Surge impedance loading (SIL) =:

A) V^2/Zc

B) V/Zc

C) Zc/V

D) $V \times Z_c$

Q.15 [2 Marks] [MODERATE] [Power Electronics — Cycloconverters]

Full-wave bridge rectifier PIV:

A) V_m
B) 2V_m
C) V_m/2
D) " 2 V_m

Q.16 [2 Marks] [MODERATE] [Control Systems — State Space]

Ideal op-amp input impedance:

A) Zero
B) Infinite
C) 1M:•
D) Output impedance

Q.17 [2 Marks] [MODERATE] [Signals & Systems — Sampling]

Slip in induction motor is:

A) Ns/Nr
B) (Ns-Nr)/Ns
C) Nr/Ns
D) (Nr-Ns)/Nr

Q.18 [2 Marks] [MODERATE] [Analog Electronics — Diodes]

DC machine with lap winding has brushes:

A) 1
B) P
C) P+1
D) 2

Q.19 [2 Marks] [MODERATE] [Electrical Machines — Servo Motors]

No-load current in transformer is called:

A) Magnetizing current
B) Excitation current
C) No-load current
D) All of the above

Q.20 [2 Marks] [MODERATE] [Power Systems — Economic Dispatch]

Surge impedance loading (SIL) =:

A) V^2/Zc
B) V/Zc
C) Zc/V
D) V×Zc

Q.21 [2 Marks] [MODERATE] [Power Electronics — Controlled Rectifiers]

Full-wave bridge rectifier PIV:

A) V_m

B) 2V_m

C) $V_m/2$

D) " 2 V_m

Q.22 [2 Marks] [MODERATE] [Control Systems — Routh-Hurwitz]

Ideal op-amp input impedance:

A) Zero
B) Infinite
C) 1M:•
D) Output impedance

Q.23 [2 Marks] [MODERATE] [Signals & Systems — Convolution]

Slip in induction motor is:

A) Ns/Nr
B) (Ns-Nr)/Ns
C) Nr/Ns
D) (Nr-Ns)/Nr

Q.24 [2 Marks] [MODERATE] [Analog Electronics — BJT/FET Amplifiers]

DC machine with lap winding has brushes:

A) 1
B) P
C) P+1
D) 2

Q.25 [2 Marks] [MODERATE] [Electrical Machines — DC Machines]

No-load current in transformer is called:

A) Magnetizing current
B) Excitation current
C) No-load current
D) All of the above

Q.26 [2 Marks] [MODERATE] [Power Systems — Transmission Lines]

Surge impedance loading (SIL) =:

A) V^2/Zc
B) V/Zc
C) Zc/V
D) V×Zc

Q.27 [2 Marks] [MODERATE] [Power Electronics — Cycloconverters]

Full-wave bridge rectifier PIV:

A) V_m
B) 2V_m
C) V_m/2
D) " 2 V_m

Q.28 [2 Marks] [MODERATE] [Control Systems — State Space]

Ideal op-amp input impedance:

A) Zero

B) Infinite

C) 1M:•

D) Output impedance

Q.29 [2 Marks] [MODERATE] [Signals & Systems — Filters]

Slip in induction motor is:

A) Ns/Nr
B) (Ns-Nr)/Ns
C) Nr/Ns
D) (Nr-Ns)/Nr

Q.30 [2 Marks] [MODERATE] [Analog Electronics — Diodes]

DC machine with lap winding has brushes:

A) 1
B) P
C) P+1
D) 2

Q.31 [2 Marks] [MODERATE] [Electrical Machines — Servo Motors]

No-load current in transformer is called:

A) Magnetizing current
B) Excitation current
C) No-load current
D) All of the above

Q.32 [2 Marks] [MODERATE] [Power Systems — Transmission Lines]

Surge impedance loading (SIL) =:

A) V^2/Zc
B) V/Zc
C) Zc/V
D) V×Zc

Q.33 [2 Marks] [HARD] [Power Electronics — Choppers]

Full-wave bridge rectifier PIV:

A) V_m
B) 2V_m
C) V_m/2
D) " 2 V_m

Q.34 [2 Marks] [HARD] [Control Systems — Root Locus]

Ideal op-amp input impedance:

A) Zero
B) Infinite
C) 1M:•
D) Output impedance

Q.35 [2 Marks] [HARD] [Signals & Systems — Sampling]

Slip in induction motor is:

A) Ns/Nr

B) (Ns-Nr)/Ns

C) Nr/Ns

D) (Nr-Ns)/Nr

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Analog Electronics — Op-Amps]

Which are rotating electrical machines?

A) Transformer
B) Induction motor
C) DC motor
D) Sync generator
E) Alternator

Q.37 [2 Marks] [HARD] [Electrical Machines — Synchronous Machines]

Which are rotating electrical machines?

A) Transformer
B) Induction motor
C) DC motor
D) Sync generator
E) Alternator

Q.38 [2 Marks] [HARD] [Power Systems — Protection]

Which are rotating electrical machines?

A) Transformer
B) Induction motor
C) DC motor
D) Sync generator
E) Alternator

Q.39 [2 Marks] [HARD] [Power Electronics — Rectifiers]

Which are rotating electrical machines?

A) Transformer
B) Induction motor
C) DC motor
D) Sync generator
E) Alternator

Q.40 [2 Marks] [HARD] [Control Systems — Compensators]

Which are rotating electrical machines?

A) Transformer
B) Induction motor
C) DC motor
D) Sync generator
E) Alternator

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Power Electronics — Rectifiers]

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: ____________________________

Q.52 [2 Marks] [HARD] [Control Systems — Compensators]

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: ____________________________

Q.53 [2 Marks] [HARD] [Signals & Systems — Sampling]

Buck converter Vin=48V, D=0.5. Vout = ?

Answer: ____________________________

Q.54 [2 Marks] [HARD] [Analog Electronics — Power Amplifiers]

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer: ____________________________

Q.55 [2 Marks] [HARD] [Electrical Machines — Induction Motors]

DC generator: 4-pole, wave wound, Z=500 conductors. Parallel paths = ?

Answer: ____________________________

Q.56 [2 Marks] [HARD] [Power Systems — Load Flow]

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: ____________________________

Q.57 [2 Marks] [HARD] [Power Electronics — Cycloconverters]

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: ____________________________

Q.58 [2 Marks] [HARD] [Control Systems — Routh-Hurwitz]

Buck converter Vin=48V, D=0.5. Vout = ?

Answer: ________________________

Q.59 [2 Marks] [HARD] [Signals & Systems — Sampling]

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer: ____________________________

Q.60 [2 Marks] [HARD] [Analog Electronics — Op-Amps]

DC generator: 4-pole, wave wound, Z=500 conductors. Parallel paths = ?

Answer: ____________________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	D	Q.2:	A	Q.3:	B
Q.4:	B	Q.5:	B	Q.6:	B
Q.7:	D	Q.8:	A	Q.9:	B
Q.10:	B	Q.11:	B	Q.12:	B
Q.13:	D	Q.14:	A	Q.15:	B
Q.16:	B	Q.17:	B	Q.18:	B
Q.19:	D	Q.20:	A	Q.21:	B
Q.22:	B	Q.23:	B	Q.24:	B
Q.25:	D	Q.26:	A	Q.27:	B
Q.28:	B	Q.29:	B	Q.30:	B
Q.31:	D	Q.32:	A	Q.33:	B
Q.34:	B	Q.35:	B	Q.36:	BCDE
Q.37:	BCDE	Q.38:	BCDE	Q.39:	BCDE
Q.40:	BCDE	Q.41:	BCDE	Q.42:	BCDE
Q.43:	BCDE	Q.44:	BCDE	Q.45:	BCDE
Q.46:	BCDE	Q.47:	BCDE	Q.48:	BCDE
Q.49:	BCDE	Q.50:	BCDE	Q.51:	121
Q.52:	1500	Q.53:	24	Q.54:	60
Q.55:	2	Q.56:	121	Q.57:	1500
Q.58:	24	Q.59:	60	Q.60:	2

Detailed Solutions

Question 1
[Electrical Machines — Transformers]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 2
[Power Systems — Economic Dispatch]

Surge impedance loading (SIL) =:

Answer: (A)

SIL = V^2/Z_c where Z_c = " (L/C) is surge impedance.

Question 3
[Power Electronics — Inverters]

Full-wave bridge rectifier PIV:

Answer: (B)

PIV = $2V_m$ in full-wave bridge rectifier.

Question 4
[Control Systems — Compensators]

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 5
[Signals & Systems — Sampling]

Slip in induction motor is:

Answer: (B)

s = (Ns - Nr)/Ns. Fraction of synchronous speed lost.

Question 6
[Analog Electronics — Op-Amps]

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 7
[Electrical Machines — DC Machines]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 8
[Power Systems — Load Flow]

Surge impedance loading (SIL) =:

Answer: (A)

SIL = V^2/Z_c where Z_c = " (L/C) is surge impedance.

Question 9
[Power Electronics — Inverters]

Full-wave bridge rectifier PIV:

Answer:

(B)

PIV = 2V_m in full-wave bridge rectifier.

Question 10 *[Control Systems — Compensators]*

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 11 *[Signals & Systems — Convolution]*

Slip in induction motor is:

Answer: (B)

s = (Ns - Nr)/Ns. Fraction of synchronous speed lost.

Question 12 *[Analog Electronics — BJT/FET Amplifiers]*

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 13 *[Electrical Machines — Servo Motors]*

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 14 *[Power Systems — Economic Dispatch]*

Surge impedance loading (SIL) =:

Answer: (A)

SIL = V^2/Zc where Zc = " (L/C) is surge impedance.

Question 15 *[Power Electronics — Cycloconverters]*

Full-wave bridge rectifier PIV:

Answer: (B)

PIV = 2V_m in full-wave bridge rectifier.

Question 16 *[Control Systems — State Space]*

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 17 *[Signals & Systems — Sampling]*

Slip in induction motor is:

Answer: (B)

s = (Ns - Nr)/Ns. Fraction of synchronous speed lost.

Question 18 *[Analog Electronics — Diodes]*

DC machine with lap winding has brushes:

Answer:

(B)

Lap winding needs P brushes for P poles.

Question 19 *[Electrical Machines — Servo Motors]*

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 20 *[Power Systems — Economic Dispatch]*

Surge impedance loading (SIL) =:

Answer: (A)

SIL = V^2/Zc where Zc = " (L/C) is surge impedance.

Question 21 *[Power Electronics — Controlled Rectifiers]*

Full-wave bridge rectifier PIV:

Answer: (B)

PIV = $2V_m$ in full-wave bridge rectifier.

Question 22 *[Control Systems — Routh-Hurwitz]*

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 23 *[Signals & Systems — Convolution]*

Slip in induction motor is:

Answer: (B)

s = (Ns - Nr)/Ns. Fraction of synchronous speed lost.

Question 24 *[Analog Electronics — BJT/FET Amplifiers]*

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 25 *[Electrical Machines — DC Machines]*

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 26 *[Power Systems — Transmission Lines]*

Surge impedance loading (SIL) =:

Answer: (A)

SIL = V^2/Zc where Zc = " (L/C) is surge impedance.

Question 27 *[Power Electronics — Cycloconverters]*

Full-wave bridge rectifier PIV:

Answer:

(B)

PIV = 2V_m in full-wave bridge rectifier.

Question 28 *[Control Systems — State Space]*

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 29 *[Signals & Systems — Filters]*

Slip in induction motor is:

Answer: (B)

s = (Ns - Nr)/Ns. Fraction of synchronous speed lost.

Question 30 *[Analog Electronics — Diodes]*

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 31 *[Electrical Machines — Servo Motors]*

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 32 *[Power Systems — Transmission Lines]*

Surge impedance loading (SIL) =:

Answer: (A)

SIL = V^2/Z_c where Z_c = " (L/C) is surge impedance.

Question 33 *[Power Electronics — Choppers]*

Full-wave bridge rectifier PIV:

Answer: (B)

PIV = 2V_m in full-wave bridge rectifier.

Question 34 *[Control Systems — Root Locus]*

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 35 *[Signals & Systems — Sampling]*

Slip in induction motor is:

Answer: (B)

s = (Ns - Nr)/Ns. Fraction of synchronous speed lost.

Question 36 *[Analog Electronics — Op-Amps]*

Which are rotating electrical machines?

Answer:

(B, C, D, E)

Transformers are static. Others are rotating machines.

Question 37 *[Electrical Machines — Synchronous Machines]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 38 *[Power Systems — Protection]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 39 *[Power Electronics — Rectifiers]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 40 *[Control Systems — Compensators]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 41 *[Signals & Systems — Sampling]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 42 *[Analog Electronics — Oscillators]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 43 *[Electrical Machines — DC Machines]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 44 *[Power Systems — Protection]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 45 *[Power Electronics — Inverters]*

Which are rotating electrical machines?

Answer:

(B, C, D, E)

Transformers are static. Others are rotating machines.

Question 46 *[Control Systems — State Space]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 47 *[Signals & Systems — Filters]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 48 *[Analog Electronics — Power Amplifiers]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 49 *[Electrical Machines — Synchronous Machines]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 50 *[Power Systems — Economic Dispatch]*

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 51 *[Power Electronics — Rectifiers]*

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: (121)

I = S/V = 50000/415 "H 120.5 "H 121 A.

Question 52 *[Control Systems — Compensators]*

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: (1500)

$N_s = 120f/P = 120 \times 50/4 = 1500$ rpm.

Question 53 *[Signals & Systems — Sampling]*

Buck converter Vin=48V, D=0.5. Vout = ?

Answer: (24)

$V_{out} = D \times V_{in} = 0.5 \times 48 = 24$V.

Question 54 *[Analog Electronics — Power Amplifiers]*

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer:

(60)

Avg = 0.6 × 100 = 60 kW.

Question 55 *[Electrical Machines — Induction Motors]*

DC generator: 4-pole, wave wound, Z=500 conductors. Parallel paths = ?

Answer: (2)

Wave winding always has 2 parallel paths regardless of poles.

Question 56 *[Power Systems — Load Flow]*

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: (121)

I = S/V = 50000/415 "H 120.5 "H 121 A.

Question 57 *[Power Electronics — Cycloconverters]*

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: (1500)

$N_s = 120f/P = 120 \times 50/4 = 1500$ rpm.

Question 58 *[Control Systems — Routh-Hurwitz]*

Buck converter Vin=48V, D=0.5. Vout = ?

Answer: (24)

Vout = D×Vin = 0.5×48 = 24V.

Question 59 *[Signals & Systems — Sampling]*

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer: (60)

Avg = 0.6 × 100 = 60 kW.

Question 60 *[Analog Electronics — Op-Amps]*

DC generator: 4-pole, wave wound, Z=500 conductors. Parallel paths = ?

Answer: (2)

Wave winding always has 2 parallel paths regardless of poles.
